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Homogeneous Catalysis

Reductive Cross-Coupling of Alcohol Derivatives with Chlorosilanes via Pyridines-Promoted Si—Cl Activation

Changpeng Chen[+], Tenggang Jiao[+], Zhitao Zhang, Xiaoming Zeng, and Xuefeng Cong**

Abstract: The development of efficient approaches mediated by silyl radicals for constructing C—Si bonds is of great interest and importance, yet methods to generate these intermediates from readily available chlorosilanes remain scarce. Herein, we report a new strategy for silyl radical generation via pyridines-promoted reductive Si—Cl activation of chlorosilanes and demonstrate its application in the reductive cross-coupling with a wide array of benzyl, allyl, and propargyl alcohol derivatives. This protocol offers an efficient and step-economical route for the synthesis of diverse benzyl-, allyl-, and allenylsilanes, featuring broad substrate scope and easy scalability. Mechanistic investigations suggest that the coupling reaction was initiated by the reductive Si—Cl activation of chlorosilanes with magnesium promoted by pyridine derivatives, which generates silyl radicals. These silyl radicals subsequently mediate the deoxygenation of alcohol derivatives to form carbon radicals, culminating in C—Si radical–radical cross-coupling.

Introduction

Organosilicon compounds have broad applications in many research fields of modern chemistry. In particular, benzyl-, allyl-, and allenylsilanes serve as versatile building blocks in synthetic chemistry and material science due to their diverse reactivity.[1–12] Therefore, the development of an efficient and general strategy for the synthesis of these organosilicon compounds is of great interest and much importance. Cross-coupling reactions with silicon reagents have emerged as one of the most attractive and powerful strategies for constructing silicon-containing molecules.[13–18] Remarkable advances have recently been made to develop cross-coupling of carbon electrophiles with diverse silicon nucleophiles by transition metal catalysts, greatly expanding access to functionalized organosilanes.[19–36] However, the reductive cross-coupling of carbon electrophiles with silicon electrophiles has remained underdeveloped to date. Given the diversity and abundance of both carbon electrophiles and silicon electrophiles, developing an efficient and general strategy for the reductive cross-coupling of these two bench-stable electrophiles to access organosilanes such as benzyl-, allyl-, and allenylsilanes is of great interest and importance.(Supporting Information)

Chlorosilanes are among the most accessible feedstock chemical in organosilicon chemistry. Recently, transition-metal-catalyzed reductive cross-coupling reactions of chlorosilanes have become a powerful tool for the C—Si bond formation,[37–40] and the coupling reactions with various carbon electrophiles have been achieved.[37–52] These coupling reactions are facilitated by oxidative addition of a Si—Cl bond to a low-valent metal center to form a silametalated intermediate (Scheme 1a). Alternatively, the direct generation of a silyl radical by means of the reductive activation of chlorosilanes followed by coupling with a carbon electrophile or radical is a synthetically appealing strategy for the construction of C—Si bonds. However, such coupling reaction remains rarely studied due to the challenging single-electron reduction of the strong Si—Cl bonds. Recently, Lin and coworkers developed an electro reductive approach to access silyl radicals by activating chlorosilanes with attractive applications in coupling with various alkenes for the formation of the C—Si bonds.[53,54] Herein, we report a new and alternative method for generating silyl radicals via the reductive activation of chlorosilanes with magnesium promoted by pyridine derivatives and demonstrate its attractive application in the reductive cross-coupling with a wide array of benzyl, allyl, and propargyl alcohol derivatives (Scheme 1b). This work represents not only a new strategy for silyl radical generation from readily available chlorosilane but also a rare example of reductive cross-coupling of alcohol derivatives with chlorosilanes without the need for transition-metal-catalysts, offering an efficient and step-economical route for the synthesis of diverse benzyl-, allyl-, and allenylsilanes.

Results and Discussion

At first, we examined the cross-coupling reaction of 2-naphthylmethyl methyl ether **1a** with chlorotrimethylsilane **2a** by using 99.99% purity magnesium as the reductant (Table 1). When the reaction was carried out in THF at 80 °C, no

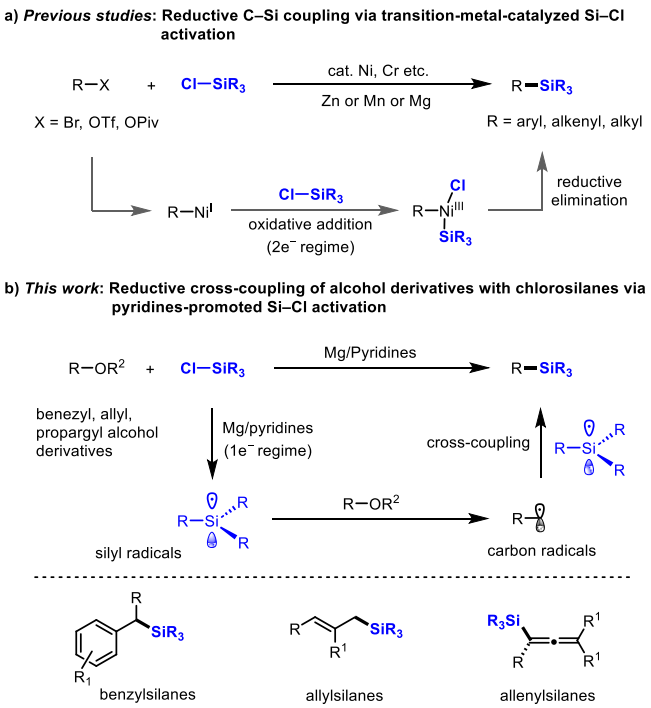
[*] C. Chen[+], T. Jiao[+], Z. Zhang, X. Cong
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E-mail: zengxiaoming@scu.edu.cn

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Additional supporting information can be found online in the Supporting Information section

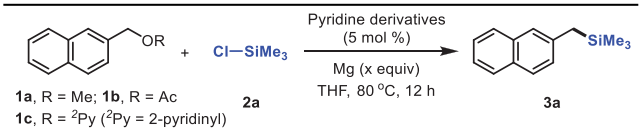
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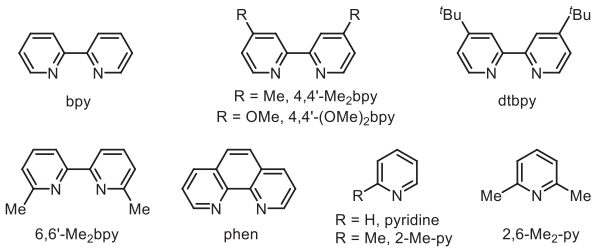
Scheme 1. Reductive cross-coupling of chlorosilanes via Si—Cl activation.

significant reaction between **1a** and **2a** was observed (Table 1, entry 1). To our delight, addition of a catalytic amount of 2,2′-bipyridine (5 mol%) into the reaction dramatically promoted the reaction, affording the cross-coupling product **3a** in 59% yield (Table 1, entry 2). The replacement of 2,2′-bipyridine by 4,4′-dimethyl-2,2′-bipyridyl further increased the yield of coupling product **3a** to 72% (Table 1, entry 3), whereas 4,4′-dimethoxyl-2,2′-dipyridyl showed a slightly lower reactivity (Table 1, entry 4). The use of 4,4′-di-*tert*-butyl-2,2′-dipyridyl (dtbpy) remarkably increased the reactivity to afford the desired coupling product **3a** in 90% yield (Table 1, entry 5). In contrast, 6,6′-di-methyl-2,2′-dipyridyl and *o*-phenanthroline showed much less efficiency in promoting the cross-coupling of **1a** with **2a** (Table 1, entries 6 and 7). It was worth noting that various pyridines such as pyridine, 2-methylpyridine and 2,6-dimethylpyridine are also effective for the present coupling reaction with slightly diminished yields (Table 1, entries 8–10). In the presence of a catalytic amount of dtbpy (5 mol%), decreasing the amount of magnesium from 2.0 equiv. to 1.0 equiv. lowered the product yield to 50% (Table 1, entry 11). No significant reaction between **1a** and **2a** was observed in the absence of a magnesium reductant (Table 1, entry 12). Subsequently, the coupling reaction of other 2-naphthalenylmethyl alcohol derivatives with **2a** was examined. The reaction of 2-naphthalenylmethyl acetate **1b** with **2a** showed a similar performance with that of **1a**, giving the desired coupling product **3a** in 75% yield in the presence of dtbpy (5 mol%) but no product **3a** in the absence of dtbpy (5 mol%) (Table 1, entries 13 and 14). In the reaction of 2-naphthylmethyl 2-pyridyl ether **1c** with **2a**, the absence of dtbpy did not influence the reactivity, affording the coupling product **3a** in 90% yield (Table 1, entries 15 and 16). These

Table 1: Optimization of reaction conditions.[a]



Entry	1	Pyridine derivatives	Mg	Yield of **3a**
1	**1a**	-	2 equiv	n.d.
2	**1a**	bpy	2 equiv	59%
3	**1a**	4,4'-Me_2bpy	2 equiv	72%
4	**1a**	4,4'-$(OMe)_2$bpy	2 equiv	46%
5	**1a**	dtbpy	2 equiv	90%
6	**1a**	6,6'-Me_2bpy	2 equiv	trace
7	**1a**	phen	2 equiv	11%
8	**1a**	pyridine	2 equiv	71%
9	**1a**	2-Me-py	2 equiv	67%
10	**1a**	2,6-Me_2-py	2 equiv	63%
11	**1a**	dtbpy	-	n.d.
12	**1a**	dtbpy	1 equiv	50%
13	**1b**	dtbpy	2 equiv	75%
14	**1b**	-	2 equiv	n.d.
15	**1c**	dtbpy	2 equiv	90%
16	**1c**	-	2 equiv	90%



[a] Reaction conditions unless otherwise noted: **1a** (0.4 mmol), **2a** (0.8 mmol), pyridine derivatives (5 mol%), Mg (99.99% purity, 0.8 mmol), THF (1 mL), 80 °C, 12 h, isolated yield.

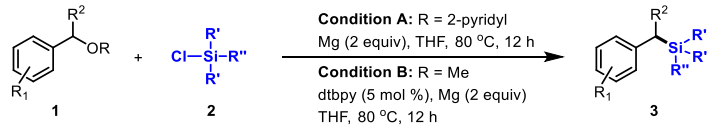
results indicated that the pyridine unit played a critically important role in promoting the cross-coupling reaction of 2-naphthylmethyl alcohol derivatives with **2a**.

Having established the cross-coupling reaction of 2-naphthylmethyl alcohol derivatives **1a–1c** with **2a**, we then investigated the coupling reactions of various (hetero)arylmethyl ethers **1** with **2**. Some representative results are summarized in Table 2. In addition to **1a**, 1-naphthylmethyl 2-pyridyl ether was also suitable substrate for the reaction with **2a**, affording the coupling product **3b** in 89% yield. Benzyl 2-pyridyl ether derivatives bearing diverse substituents at *para*-, *meta*-, or *ortho*-position of the benzyl group all worked well with **2a**, affording the corresponding coupling products **3c–3ae** in 46%–87% yields. Function groups such as OPh (**3d**, **3j**), OBn (**3s**, **3z**), F (**3g**, **3v**), NMe_2 (**3k**), CF_3 (**3m**), OCF_3 (**3x**), $SiMe_3$ (**3p**), and aromatic heterocycles (**3t**, **3u**) were all well-tolerated. Interestingly, the two C–O bonds of 1,4-bis((pyridin-2-yloxy)methyl)benzene were coupled with **2a** to form the bis-silylation product **3af** in 53% yield. The reaction of 4-pyridylmethyl 2-pyridyl ether gave the corresponding product **3ag** in 66% yield. Notably,

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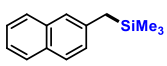
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Table 2: Reductive cross-coupling reactions of various (hetero)arylmethyl ether derivatives **1** with **2**.[a)]

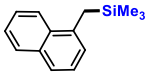


1 **2** **3**

Scope of benzyl ethers



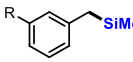
3a
90% (**A**); 90% (**B**)



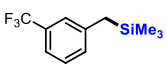
3b
89% (**A**); 80% (**B**)



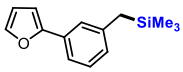
3c, R = Me, 67% (**A**); 37% (**B**)
3d, R = OPh, 72% (**A**); 70% (**B**)
3e, R = Ph, 71% (**A**); 76% (**B**)
3f, R = *p*-toyl, 72% (**A**); 80% (**B**)
3g, R = F, 49%[b)] (**A**)



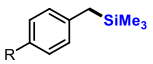
3h, R = Me, 65% (**A**); 66% (**B**)
3i, R = OMe, 61%[b)] (**A**); 46% (**B**)
3j, R = OPh, 66% (**A**); 70% (**B**)
3k, R = NMe_2, 49% (**A**); 50% (**B**)
3l, R = Ph, 70% (**A**); 80% (**B**)



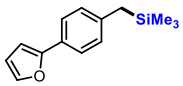
3m
77% (**A**); 64% (**B**)



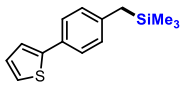
3n
66% (**A**); 70% (**B**)



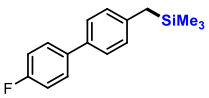
3o, R = Me, 46% (**A**); 39% (**B**)
3p, R = $SiMe_3$, 46% (**A**); 50% (**B**)
3q, R = ^{t}Bu, 77% (**A**); 70% (**B**)
3r, R = Ph, 84% (**A**); 92% (**B**)
3s, R = OBn, 56% (**A**); 60% (**B**)



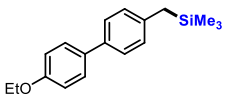
3t
56%, (**A**); 61% (**B**)



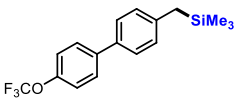
3u
83% (**A**); 88% (**B**)



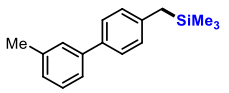
3v
80% (**A**); 88% (**B**)



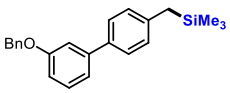
3w
42% (**A**); 82% (**B**)



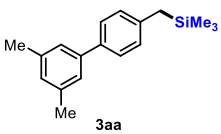
3x
85% (**A**); 89% (**B**)



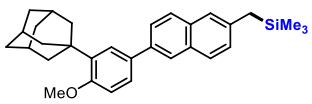
3y
79% (**A**); 90% (**B**)



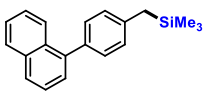
3z
86% (**A**); 90% (**B**)



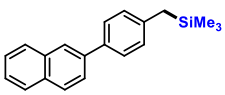
3aa
85% (**A**); 91% (**B**)



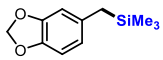
3ab
82% (**A**); 81% (**B**)



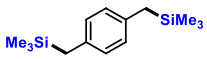
3ac
81% (**A**); 83% (**B**)



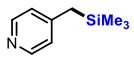
3ad
87% (**A**); 90% (**B**)



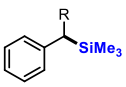
3ae
51%[c)] (**A**); 70%[c)] (**B**)



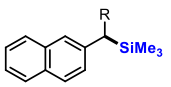
3af, 53%[d)] (**A**)



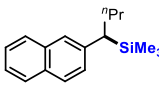
3ag, 66% (**A**)



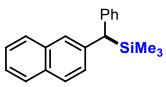
3ah, R = Bn, 35% (**A**)
3ai, R = Ph, 71% (**A**)



3aj, **R =** Me, 45% (**A**); 72% (**B**)
3ak, **R =** Et, 55% (**A**); 75% (**B**)

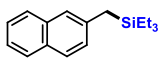


3al
76% (**A**); 75% (**B**)

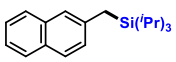


3am
50% (**A**); 70% (**B**)

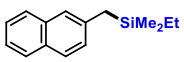
Scope of chlorosilanes



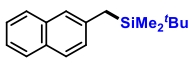
3an
82% (**A**); 90% (**B**)



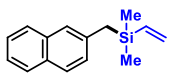
3ao
10% (**A**); 82% (**B**)



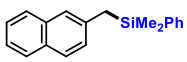
3ap
82% (**A**); 91% (**B**)



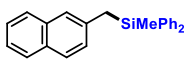
3aq
10% (**A**); 50% (**B**)



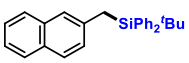
3ar
61% (**A**); 70% (**B**)



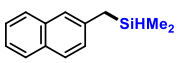
3as
79% (**A**); 82% (**B**)



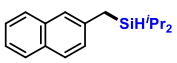
3at
80% (**A**); 71% (**B**)



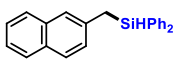
3au
trace (**A**); 39% (**B**)



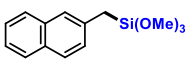
3av
82% (**A**); 62% (**B**)



3aw
74% (**A**); 81% (**B**)

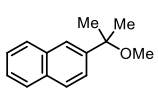


3ax
72% (**A**); 80% (**B**)

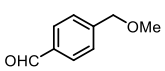


3ay
trace (**A**); trace (**B**)

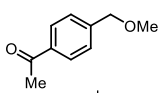
Unsuccessful benzyl ethers



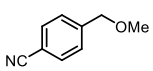
< 10%



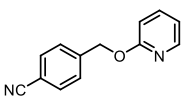
n.d.



n.d.



n.d.



n.d.

[a)] Reaction conditions unless otherwise noted: **Method A: 1** (0.4 mmol), **2** (0.8 mmol), Mg (99.99% purity, 0.8 mmol), THF (1 mL), 80 °C, 12 h; **Method B: 1** (0.4 mmol), **2** (0.8 mmol), dtbpy (5 mol%), Mg (99.99% purity, 0.8 mmol), THF (1 mL), 80 °C, 12 h. Isolated yields are given. [b)] **2a** (1.2 mmol). [c)] 100 °C. [d)] **2a** (1.6 mmol), Mg (99.99% purity, 1.6 mmol).

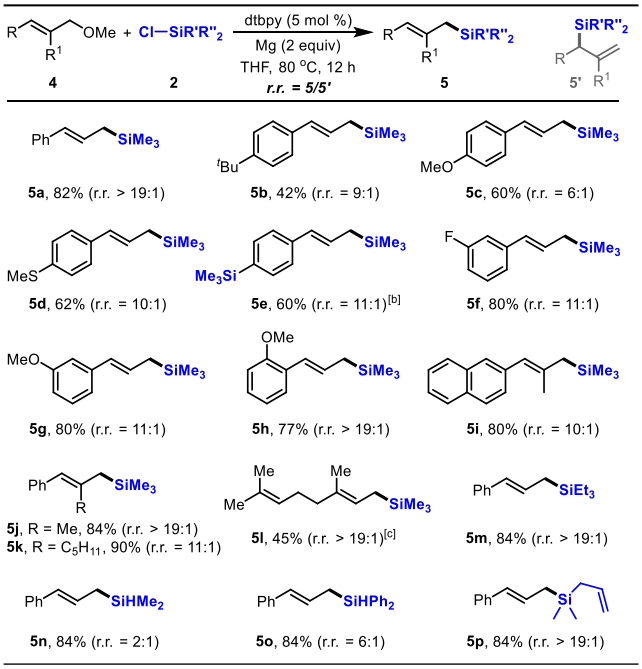
secondary benzyl and 2-naphthylmethyl ethers bearing Bn, Me, Et, ^{n}Pr, and Ph groups at the benzylic position were also compatible, affording the desired products **3ah–3am** in 35%–76% yields.

The cross-coupling reactions of various arylmethyl methyl ethers **1** with **2a** promoted by a catalytic amount of dtbpy (5 mol%) were also examined. As in the reactions of arylmethyl 2-pyridyl ethers with **2a**, the cross-coupling reactions of arylmethyl methyl ethers **1** with **2a** in the presence of a catalytic amount of dtbpy (5 mol%) were also compatible with diverse functional groups such as alkyl, aryl, $SiMe_3$, OMe, CF_3, F, OCF_3, and aromatic heterocycles. Remarkably,

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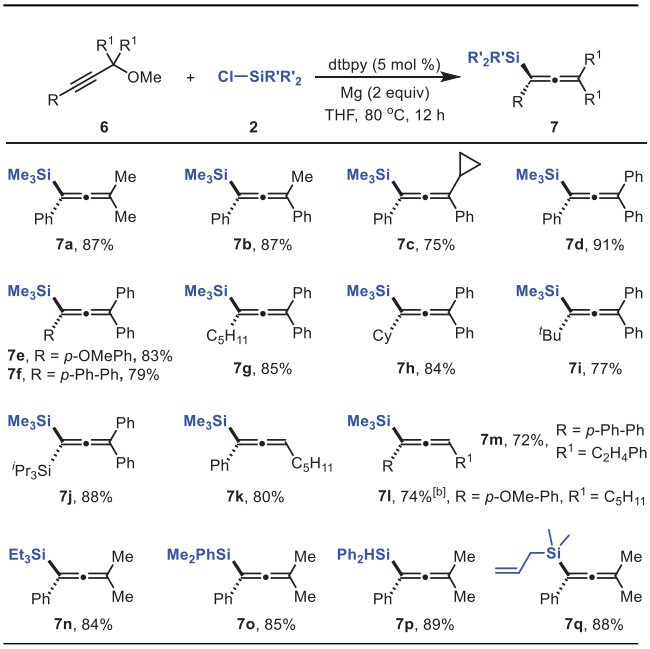
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Table 3: Reductive cross-coupling of allyl methyl ethers **4** with chlorosilanes **2** promoted by a catalytic amount of dtbpy.[a)]



[a)] Reaction conditions unless otherwise noted: **4** (0.4 mmol), **5** (0.8 mmol), dtbpy (5 mol%), Mg (99.99% purity, 0.8 mmol), THF (1 mL), 80 °C, 12 h. Isolated yields are given. [b)] 4-Chlorocinnamyl methyl ether was used. [c)] Geraniol-derived 2-pyrindyl ether (0.4 mmol) was used in the absence of dtbpy, 100 °C.

Table 4: Reductive cross-coupling of propargyl methyl ethers **6** with chlorosilanes **2** promoted by a catalytic amount of dtbpy.[a)]



[a)] Reaction conditions unless otherwise noted: **6** (0.4 mmol), **2** (0.8 mmol), dtbpy (5 mol%), Mg (99.99% purity, 0.8 mmol), THF (1 mL), 80 °C, 12 h. Isolated yields are given. [b)] Formation of the corresponding propargylsilane **7l'** was observed (**7l/7l'** = 2:1).

secondary 2-naphthylmethyl methyl ethers bearing Me, Et, ^{n}Pr, and Ph groups at the benzylic position underwent the coupling reaction with **2a** well, affording the desired products **3aj–3am** in good yields. However, this coupling reaction was incompatible with tertiary benzyl ether as well as substrates containing aldehyde, ketone, or cyano functional groups.

We then examined the cross-coupling reactions of various chlorosilanes with **1a** and **1c**, respectively. A diverse array of chlorosilanes proved to be effective for the coupling reactions with **1a** and **1c** under the optimal conditions, affording the targeted products **3am–3aw** in good yields. It was worth noting that, in the case of chlorohydrosilanes, the strong Si—Cl bonds selectively couple with benzyl C—O bonds smoothly to afford the 2-naphthylmethyl-substituted hydrosilanes **3au–3aw** in 62%–82% yields, and the reactive Si–H group remained intact. However, the reactions of chloro(trimethoxy)silane with either **1a** or **1c** were unsuccessful probably due to the electronic effect.

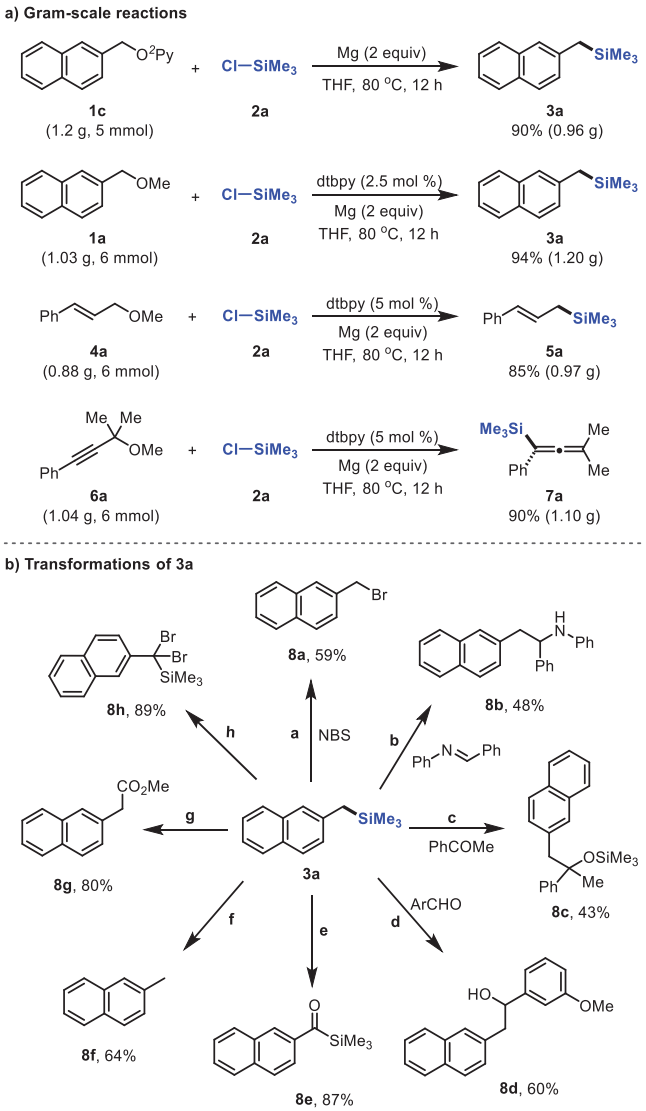
The analogous coupling reactions of allyl methyl ethers **4** with chlorosilanes **2** were then examined. As shown in Table 3, a range of allyl methyl ethers worked well with **2a** in the presence of a catalytic amount of dtbpy (5 mol%) and 2 equiv. of magnesium at 80 °C in 12 h, affording the corresponding allylsilanes **5a–5h** in 42%–82% yields with high stereoselectivity. The α-substituents such as methyl (**5i**, **5j**) and *n*-pentyl (**5k**) did not hamper the reaction with **2a**. Geraniol-derived 2-pyrindyl ether was also suitable for the reaction with **2a**, affording the corresponding product **5l** in 45% yield. In addition to **2a**, chlorotriethyl-, chlorodimethyl-, chlorodiphenyl-, and allylchlorodimethylsilanes were also compatible for the coupling reactions with **4a**, giving the targeted products **5m–5p** in 84% yields.

Table 4 summarizes the analogous coupling reactions of propargyl methyl ethers **6** with **2** in the presence of a catalytic amount of dtbpy (5 mol%). A series of propargyl methyl ethers bearing diverse alkyl and aryl groups on the alkynyl unit or at propargyl position underwent the coupling reaction with **2a** smoothly, giving the corresponding tetra- or trisubstituted silylallenes **7a–7m** in good to excellent yields (72%–91%). Triisopropylsilyl (TIPS) group directly bonded to the alkynyl unit was also compatible, affording the disilyl-substituted allene **7j** in 88% yield. In addition to **2a**, chlorotriethyl-, chlorodimethyl-, chlorodiphenyl-, and allylchlorodimethylsilanes also worked well with **6a**, affording the corresponding products **7n–7q** in 84%–89% yields.

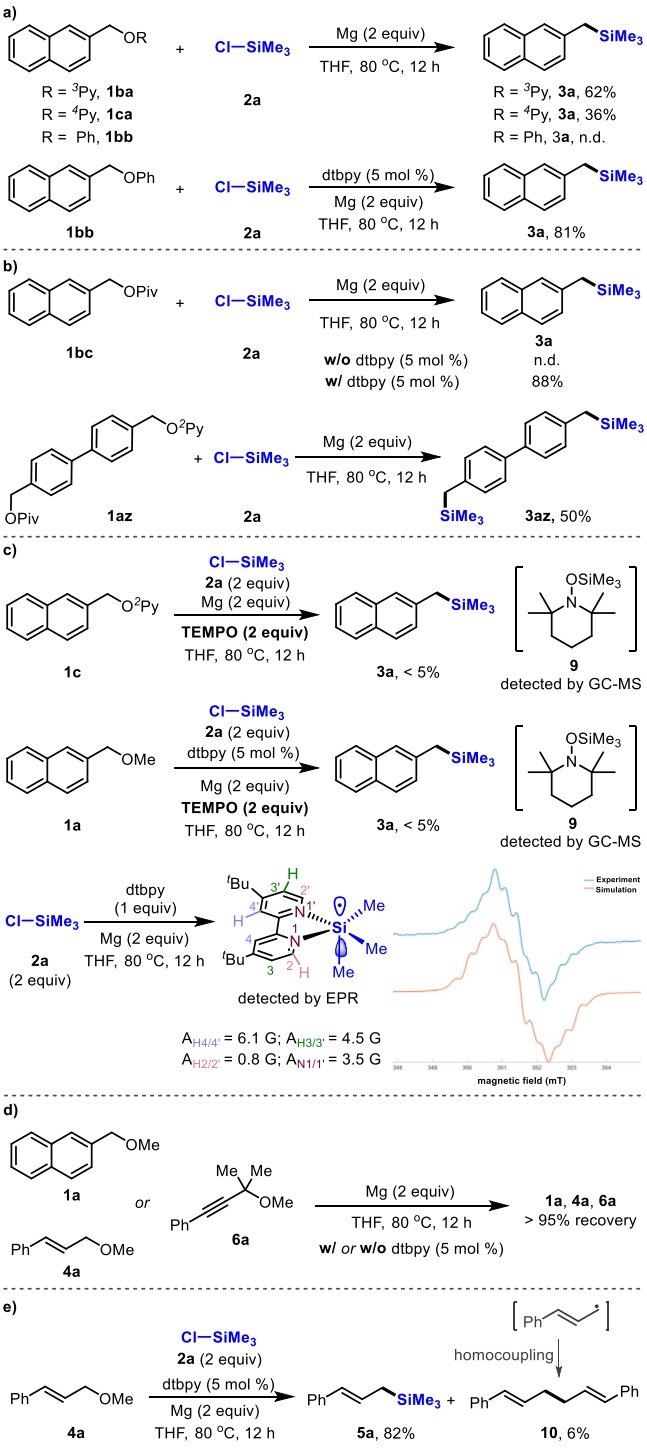
To highlight the practical value of the present protocol, some gram-scale reactions were conducted. As shown in Scheme 2a, the reactions of 5 mmol of **1a** and 6 mmol of **1c** with **2a** under the standard conditions afforded the desired product **3a** in 90% and 94% yields, respectively. Similarly, the reactions of 6 mmol of **4a** and **6a** with **2a** in the presence of a catalytic amount of dtbpy (5 mol%) yielded the corresponding allylsilane **5a** and allenylsilane **7a** in 85% and 90% yields, respectively. Scheme 2b shows some transformations of **3a**,[55–60] which afforded a variety of structurally diverse compounds including 2-(bromomethyl)naphthalene (**8a**),[55] secondary amine (**8b**),[56] silicon ether (**8c**),[57] secondary alcohol (**8d**),[57] acyl silane (**8e**),[58] 2-methylnaphthalene

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Scheme 2. a) Gram scale reactions; b) Transformations of **3a**, reaction conditions: **3a** (0.2 mmol), a) NBS (0.2 mmol), MeCN (1 mL), hν, r.t., 6 h; b) (*E*)-*N*,1-diphenylmethanimine (0.15 mmol), TBAF (5 mol%), 4 Å MS (100 mg), THF (1 mL), 80 °C, 12 h; c) Acetophenone (0.4 mmol), TBAT (5 mol%), THF (1 mL), 80 °C, 6 h; d) 3-Methoxybenzaldehyde (0.4 mmol), TBAT (5 mol%), THF (1 mL), 80 °C, 6 h; e) (i) NBS (0.4 mmol), AIBN (5 mol%), benzene (1 mL), 80 °C, 12 h; (ii) AgOAc (0.4 mmol), H_2O:Acetone:EtOH = 1:2:3 (2 mL), r.t., 24 h; f) $KOSiMe_3$ (5 mol%), DMSO (1 mL), 70 °C, 6 h; g) (i) CO_2 (1 atm), CsF (0.6 mmol), DMF (1 mL), 100 °C, 4 h, (ii) MeI (2 equiv.), Cs_2CO_3 (2 equiv.), r.t., 30 min; h) NBS (0.4 mmol), AIBN (5 mol%), benzene (1 mL), 80 °C, 12 h.



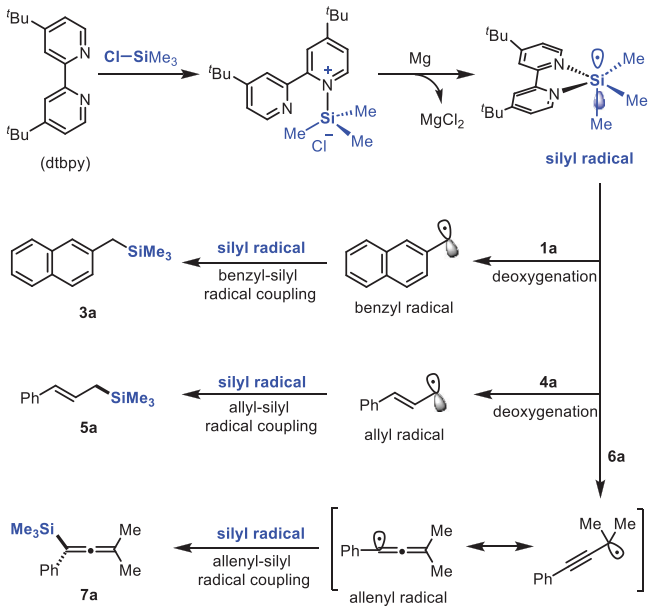
Scheme 3. Mechanistic studies.

(**8f**),[59] methyl 2-(naphthalen-2-yl)acetate (**8g**),[60] and (dibromo(naphthalen-2-yl)methyl)trimethylsilane (**8h**).[58]

To gain information on the reaction mechanism, several control experiments were carried out (Scheme 3). 2-Naphthylmethyl 3-pyrindyl ether **1ba** and 4-pyrindyl ether **1ca** reacted well with **2a**, giving the product **3a** in 62% and 36% yields, respectively. The reaction of the analogous substrate 2-naphthylmethyl phenyl ether **1bb** with **2a** did not occur, but worked well in the presence of a catalytic amount of dtbpy (5 mol%) to afford **3a** in 81% yield (Scheme 3a). Similar with that of 2-naphthalenylmethyl acetate **1b**, no desired coupling product **3a** was observed in the reaction of 2-naphthalenylmethyl pivalate **1bc** with **2a** in the absence of dtbpy (5 mol%). Interestingly, both 2-pyrindyl ether and pivalate unit of the diol derivative **1az** coupled with **2a** under the same reaction conditions, affording the disilylated product **3az** in 50% yield (Scheme 3b). These findings further indicated that the pyridine unit played a

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Scheme 4. Possible mechanism for the reductive cross-coupling of methyl ethers **1a**, **4a**, and **6a** with **2a** promoted by a catalytic amount of dtbpy.

critically important role in promoting the reductive cross-coupling of benzyl C─O bonds with Si─Cl bonds. The radical trapping experiments using 2,2,6,6-tetramethylpiperidinyl-1-oxide (TEMPO) significantly suppressed the cross-coupling of **1a** and **1c** with **2a**, and the corresponding silylated TEMPO **9** was detected by gas chromatography–mass spectrometry (GC–MS) analysis (Scheme 3c). Furthermore, the electron paramagnetic resonance (EPR) spectroscopy of the reaction mixture of dtbpy, **2a** and magnesium gave a distinct signal, which is attributed to the formation of a dtbpy-coordinated silyl radical (Scheme 3c), whereas no signal was detected for the reaction mixture of **2a** and magnesium in the absence of dtbpy. It was worth noting that no significant reaction was observed when treating **1a, 4a**, and **6a** with magnesium in the presence/absence of a catalytic amount of dtbpy (5 mol%), and **1a, 4a**, and **6a** were recovered in >95% yield (Scheme 3d). In the reaction of **4a** with **2a**, homo-coupling side-product **10** was obtained in 6% yield (Scheme 3e), indicating the potential generation of an allyl radical, which may undergo radical-radical homocoupling to form **10**.

Although more studies are required to establish a precise mechanism, based on the above experimental results and previous reports,[61–64] possible mechanisms for the reductive cross-coupling reactions of benzyl methyl ether **1a**, cinnamyl methyl ether **4a**, and propargyl methyl ether **6a** with **2a** in the presence of dtbpy are proposed as shown in Scheme 4. The combination of **2a** and dtbpy would form a siliconium-chloride salt,[65–67] which undergoes reductive activation with magnesium to generate a dtbpy-coordinated silyl radical. The subsequent deoxygenation of **1a** and **4a** by the resulting silyl radical would yield the corresponding benzyl and allyl radical, respectively.[68] The C─Si radical coupling of the benzyl and allyl radical with the silyl radical would give the desired products **3a** and **5a**, respectively, and release dtbpy. In the case of propargyl methyl ether **6a**, the deoxygenation with the silyl radical would yield a propargyl radical and its resonance hybrid allenyl radical. The allenyl–silyl radical coupling would selectively take place due to the steric reason, giving the allenylsilane **7a**. It should be noted that, although silyl radical formation was unambiguously established, a competing silyl anion pathway cannot be fully excluded.

Conclusion

In summary, we have developed a pyridines-promoted strategy for the reductive Si─Cl activation of readily available chlorosilanes and demonstrated its attractive application in the reductive cross-coupling with a wide array of benzyl, allyl, and propargyl alcohol derivatives under simple and transition-metal-free conditions. Mechanistic studies revealed that the coupling reaction proceeded through the generation of silyl radicals via pyridines-promoted reductive activation of the strong Si–Cl bond of chlorosilanes with magnesium followed by the deoxygenation of alcohol derivatives by the resulting silyl radicals to form carbon radicals and subsequent C–Si radical-radical cross-coupling. This work not only constitutes an efficient and step-economical protocol for the synthesis of diverse benzyl-, allyl,- and allenylsilanes but also offers a new and general strategy for developing new silyl radical chemistry. Further studies on transformations of chlorosilanes via the reductive activation of strong Si─Cl bonds are in progress.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Chlorosilanes · Magnesium · Pyridines · Reductive cross-coupling · Silyl radicals

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